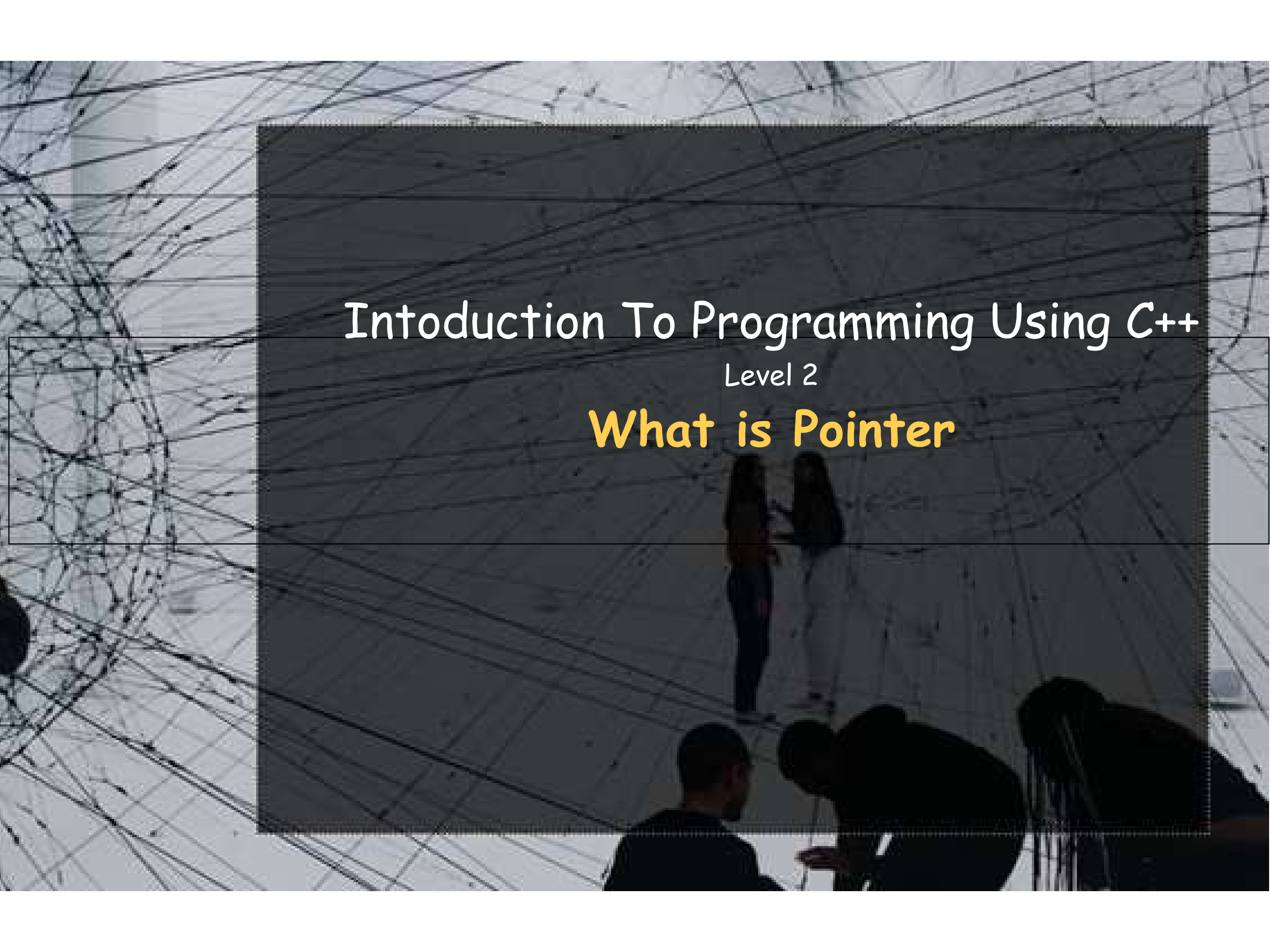


Intoduction To Programming Using C++ Level 2

What is Pointer



Introduction To Programming Using C++

Level 2

What is Pointer

Definition

Pointer

 A Pointer is a variable that stores the memory address of another variable.

 Instead of storing the actual value of another variable, a pointer stores its address in memory.

 A pointer allows us to access and modify the value stored at that memory address.

Explanation

Pointers in C++

-  Pointers are an important concept in C++ because they allow direct interaction with memory.

-  In C and C++, pointers can be used explicitly to access memory addresses.

-  This gives the programmer more control and flexibility, but it also requires careful use.

Declaring a Pointer

-  A pointer is declared using the * symbol.

-  The pointer type should match the type of variable whose address it stores.

Example:

```
int* p;
```

-  This declares p as a pointer to an int.

-  At this point, p does not point to a valid variable yet.

Getting an Address

-  The & operator is used to get the memory address of a variable.

Procedures

1. Create a variable.
2. Assign a value to the variable.
3. Declare a pointer of the same type.
4. Use & to get the address of the variable.
5. Store the address inside the pointer.
6. Use the pointer to access the value using *.
7. The pointer can later be changed to point to another variable.

💡 Example

◆ Suppose we have:

◆ Basic Pointer Example

```
int main()
{
    int a = 10;
    cout << "a value = " << a << endl;
    cout << "a address = " << &a << endl;
    int* p;
    p = &a;
    cout << "Pointer Value = " << p << endl;
    return 0;
}
```

➡ Result

a value = 10

a address = 0x61fef8

Pointer Value = 0x61fef8

Note

Pointer Types

 The pointer type should be compatible with the type of variable whose address it stores.

Example:

```
int a = 10;
```

```
int* p = &a;
```

 This is correct because p is a pointer to int.

Incorrect Example:

```
float a = 10.5;
```

```
int* p = &a;
```

 An int pointer should not be used to store the address of a float variable.

Pointer Declaration Without Initialization

 Declaring a pointer does not automatically make it point to a valid memory location.

Example:

```
int* p;
```

 At this point, p does not point to a valid object.

⊘ Warning

■ Do not use an uninitialized pointer.

▨ Incorrect:

```
int* p;  
*p = 10;
```

▨ p does not point to a valid variable, so dereferencing it causes undefined behavior.

■ Correct:

```
int a = 10;  
int* p = &a;
```

■ Always make sure that a pointer contains a valid address before dereferencing it.

✦✦ Summary

- ✓ A Pointer is a variable that stores the memory address of another variable.
- ✓ The & operator is used to get the address of a variable.
- ✓ The * operator is used to declare a pointer and to dereference it to access the value at the stored address.
- ✓ A pointer can be changed during program execution to point to another compatible variable.
- ✓ Pointers provide powerful control over memory, but they must be used carefully to avoid errors such as accessing invalid memory.

Thank you 😊